



Automated excavators activity recognition and productivity analysis from construction site surveillance videos

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ABSTRACT

The productivity of construction equipment plays an important role in completing construction projects within schedule and under budget. However, the current productivity monitoring on construction sites highly depends on manually observing and recording equipment activities, which is labor-intensive and time-consuming. To address this problem, an increasing number of research studies focused on automatically identifying equipment activities from site surveillance videos. However, these studies failed to accurately conduct the activity recognition and productivity analysis, when multiple pieces of equipment are working together. This research proposes a novel framework for automatically analyzing the activity and productivity of multiple excavators. In this framework, three convolutional neural networks are designed to detect, track and recognize the activities of excavators. The results are further compiled to analyze excavator's activity time, working cycle, and productivity. The proposed framework has been tested with the videos recorded from real construction sites. The overall activity recognition has achieved 87.6% accuracy. The productivity calculation has achieved 83% accuracy, which indicates the feasibility of the proposed framework for automating the monitoring of excavator's productivity.

1. Introduction

Construction equipment is the main component of construction production, and the equipment costs are usually one of the constructor's biggest expenditures [1]. Therefore, effectively managing equipment operations is critical to improve the construction productivity and control the cost. The systematic assessment and analysis of equipment activities information can help the construction managers to optimize equipment's operation time, improve working efficiency, and make wise project-related decisions. However, measuring and analyzing equipment operations is time-consuming, expensive, and error-prone work as it requires the site superintendents to manually observe and record the entire operation process for each construction equipment on the site [1,2]. In addition, the data analysis is also labor intensive. Therefore, an automatic method is in great need to accurately and efficiently recognize and analyze equipment activities.

Existing methods for automatically recognizing the activities of construction equipment can be classified into non-vision-based methods and vision-based methods. The non-vision-based methods categorize equipment activities by analyzing its acceleration, velocity and

orientation information from sensors, such as the Global Positioning System (GPS), inertial measurement units (IMU), etc. [3–6]. However, they require attaching sensors to monitor the equipment, which may not be feasible for the rented or old equipment [7]. Additionally, the non-vision-based methods have difficulty in categorizing the activities in detail. For example, the GPS-based methods identified equipment activities based on its position changes; and therefore, they cannot precisely identify the activity when the equipment is located at the same position while performing different activities.

Compared with the non-vision-based methods, the vision-based methods do not have the aforementioned drawbacks. Furthermore, they can visualize the equipment state directly from images and videos, so that it becomes easy to identify false recognitions and analyze the reasons behind low productivity. For example, the increase in the excavator's swinging time may be due to avoiding passing workers or the change of truck positions. Moreover, the productivity analysis and optimization could be made with the surrounding information provided by videos.

So far, the vision-based methods recognize the activities by analyzing the motion trajectories or the equipment location information in

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video frames [8]. They cannot classify the activities into detailed categories (e.g. digging, hauling, dumping, etc.) in long video sequences. In addition, most of existing methods are not able to recognize and analyze the activities when multiple pieces of equipment are captured simultaneously. Moreover, existing methods just focus on activity recognition. Their practical values are not well discussed. As a result, it is not clear how these methods can be used for productivity control in construction projects.

The main objective of this paper is to propose a framework to automatically recognize activities and analyze the productivity of multiple excavators. Excavators are selected because they are the most common earthmoving equipment, and they have various types of activities during their working process. In this framework, an excavator detector and a tracker are first integrated to identify and locate the excavators in video frames. Based on the changes of the centroid coordinates and areas of the bounding boxes of the excavators in consecutive frames, idling states are identified and the corresponding frames are cut from the video. Then, a three-dimensional convolutional neural network (3D CNN) is used to precisely recognize the excavators' activities and classify them into detailed types (e.g. digging, loading and swinging). Finally, an algorithm is developed to analyze the activity recognition results and calculate the productivity of the excavators. The feasibility of the proposed framework has been tested and validated on the construction surveillance videos collected from real construction sites.

2. Background and related works

This section first introduces existing research works focusing on construction equipment activity recognition and productivity analysis. Then, the recent developments in activity recognition in the computer vision field are also reviewed, considering that these state-of-the-art methods rely on deep learning technologies and could significantly contribute to activity recognition in construction.

2.1. Equipment activity recognition and productivity analysis

2.1.1. Non-vision-based methods

In the non-vision-based methods, different types of sensors, such as GPS, radio frequency identification (RFID) and IMU are used to collect the location and movement data of the target equipment in order to estimate its activities. GPS is a well-known tracking technology, which can provide location and trajectory information of the equipment in real time. Combined with the geographic information, it can estimate the operation and productivity of the equipment [3,4]. The RFID sensor system consists of two main components: tags and readers. The tags are installed on the equipment to transmit signals to the readers. Typically, the readers are installed in different work zones of a construction site, such as loading and dumping areas. The time of each activity is identified depending on how long the reader could receive the signals from the specific work zones [9,10]. These GPS and RFID based methods cannot precisely recognize equipment's activities, since they are not able to confirm whether the equipment is actually loading or idling in the loading area. The methods with the reliance on IMU sensors, could solve this problem. For example, in order to identify the activity of a truck and a loader, Akhavan and Behzadan [5,6] first identified the working routes of the truck and the loader. Then, they installed the IMU sensors on the bed of the truck and the boom of the loader to get the angles of the bed and boom. By analyzing the change of the angles, they could identify the equipment's activities. However, these non-vision-based methods [3–6] may have limited use in real construction projects because the sensor system could be difficult to install on the rented and old equipment. In addition, the installation and disassembly of sensor systems also requires labor and cost [11]. This is especially true for the RFID sensor systems, which need comprehensive infrastructure to be installed on both equipment and jobsite [12]. Moreover, construction sites have complex environments with noise, which may disturb or

obstruct the wireless signal transmission and decrease the accuracy of the activity recognition results.

2.1.2. Vision-based methods

Surveillance cameras have been widely installed on construction sites, considering their great potential for monitoring the daily activities of equipment and workforce [13]. These cameras are usually installed at high positions to avoid occlusions [11]. Compared with sensory data, the videos collected from the cameras can also be used for equipment activity recognition with computer vision techniques, which might be more cost-effective and applicable in real construction projects. Vision-based methods can be classified as motion-feature-based methods and detection-tracking-based methods. Currently, numerous vision-based methods have been developed for construction equipment's activity recognition and achieved promising performance.

The motion-feature-based methods first extract features from consecutive frames in a video, which represent the spatial and temporal information of an activity. Then, the features are encoded to fixed-length vectors for the activity classification. For example, Gong et al. [14] used the 3D-Harris and local histograms to extract and represent the features of the excavator's activities. Then, a Bayesian network was used as the learning mechanism to classify activities into three types: relocating, excavating, and swinging. The activity recognition schema proposed by Golparvar-Fard et al. [15] was similar to the one proposed by Gong et al. [14]. The major difference is that they used the 3D Histograms of Oriented Gradients (HOG) to represent the motion features and a support vector machine (SVM) classifier to recognize the digging, hauling, dumping and swinging of the excavator. To improve the recognition performance, Yang et al. [8] applied three types of feature representations, HOG, Histogram of Optical Flow (HOF) and Motion Boundary Histograms (MBH) to describe the motion information of an activity, and used an SVM classifier to identify it. These studies have been tested to recognize activities of the construction equipment and achieved the recognition accuracy ranging from 73.6% to 86.3%. One major limitation of the motion-feature-based methods is that the motion trajectories of the features always drift with the progress of the time. In addition, in these methods, the start and finish times of each activity in video sequences have to be pre-defined [15]. Therefore, it is difficult for them to recognize consecutive activities in long video sequences.

The detection-tracking-based methods mainly rely on object detection and tracking methods to identify the equipment activities and working cycles in long videos. They first get the equipment locations in videos from detection and tracking results. Then, the types of activities are recognized by comparing the equipment spatial relationships, as well as their location changes in consecutive video frames. Zou and Kim [16] used the color space to track the excavator and get its centroid coordinates. By comparing the changes of the centroid points in consecutive frames, they could differentiate the working and stopping activities of an excavator and estimate its productivity. Bügler et al. [17] separated a construction site to several regions that needed earthwork. Then, they used the Gaussian mixture model (GMM) method [18] to detect the excavator and the truck. If the excavator and truck appeared in the same region of interest, their activities, such as stopping, filling and moving, are estimated. Similarly, Kim et al. [19] used an HOG-based detector to get the locations of the excavator and the truck, and identified their activities with the centroid coordinates changing in consecutive frames. In order to achieve more accurate results, they adopted an interaction analysis to determine whether the excavator and the truck work together. For example, only if the truck is static and the excavator is working, the activity of the excavator could be classified as actual work. However, the accuracy of these detection-tracking-based methods [16,17,19] is not sufficient for productivity calculation. They could just identify working, idling and stopping activities, which is not enough to determine excavator's working cycles.

The excavator's productivity is calculated based on the number of

cycles it performs per unit of time. Therefore, in order to achieve an accurate result, other research studies focused on determining the working cycles of the equipment. Azar et al. [11] used two HOG-based detectors to get the locations of the excavator and the truck. Then, they trained an SVM classifier to identify the loading activity using the vectors computed from the distance between the base point of the excavator and four corners of the dump truck. The time between two loading activities was identified as a loading cycle, and then the productivity was calculated. Kim et al. [20] created a more accurate method to define the cycle time. They first used a CNN to detect the excavator and the truck. Then, the loading activity is identified when the centroid distance between the excavator and the truck starts changing within a certain threshold. Luo et al. [21] presented a method to recognize diverse construction interaction activities based on objects' semantic and spatial relevance with each other. In their work, the object detection method Faster Region-based Convolutional Neural Network (Faster R-CNN) [22] was used to detect 22 categories of construction-related objects, such as equipment, materials and workers. Then, relevance networks of the objects were created based on the semantic and spatial relationship between these construction objects to determine activities of patterns in the network. In this work, 20 activities have been recognized with the accuracy of 86.33%.

The most recent research of activity recognition in construction was built on hybrid methods. Kim et al. [23] used Faster R-CNN detector and detection-based tracker to get the locations of the excavator in video frames. Then they combined CNN and the Long Short Term Memory (LSTM) network to recognize digging, dumping and hauling activities of the excavator based on their sequential relationship. Although their activity recognition accuracy reached 90.9%, their method could not conduct activity recognition in long video sequences because of the gradient descent problem in LSTM. In their paper, Kim et al. [23] indicated that the sequential activities were recognized in a short video of about 11 s (116 frames with 10 frames per second). However, monitoring the excavator's operation in long video sequences is essential for productivity analysis. In addition, since activities are identified according to their sequence relations in the method of Kim et al. [23], their method cannot accurately identify out-of-sequence activities.

2.2. State-of-the-art activity recognition in computer vision

In recent years, video activity recognition has made great progress with the development of deep learning methods in computer vision. The deep learning methods could achieve high accuracy in activity recognition. Furthermore, they are much simpler to use than the traditional feature-based methods since the representation of the spatial and temporal features can be automatically learned by the CNN. There are two main approaches in video activity recognition with deep neural networks: two-stream methods and deep 3D CNN methods.

Simonyan and Zisserman [24] first proposed the two-stream method. In their method, two CNNs were used to extract the special and temporal information from the frames and optical flows of a video separately. The spatial stream CNN took single RGB frames as input to extract the appearance representation of the activity in videos; and the temporal stream CNN, with the same structure as the spatial one, took several optical flows as input to describe the temporal features between consecutive video frames. At last, the two CNNs were fused by averaging the softmax scores to provide the final activity recognition results. The result showed the method of Simonyan and Zisserman [24] achieved the accuracy of 88% on testing human activities.

Following that, numerous variations and improvements have been made. Wang et al. [25] applied the data augmentation, multi-GPU, pre-training and smaller learning rates techniques to train deep two-stream CNNs, which achieved better activity recognition accuracy. Ng et al. [26] improved the activity recognition performance on long videos by combining the widely used GoogLeNet with the LSTM. Wang [27] presented a more efficient Temporal Segment Network (TSN). Instead

of operating all the frames in the videos, the TSN worked on a sequence of short snippets sparsely sampled from an input video, so that it could efficiently capture the long-term temporal features of the video. Zhu et al. [28] presented a hidden two-stream method, which used a fully convolutional neural network to implicitly capture the temporal information between consecutive frames instead of explicitly computing the optical flow. The experimental results showed that the hidden two-stream method was ten times faster than the original two-stream method. The novel two-stream method has already been used in construction workers' activities to monitor the labor productivity. Luo et al. [29,30] used TSN [27] to recognize the workers' activities into 16 detailed categories by taking both RGB and optical flow images as input. The recognition accuracy achieved 80.5%. Although the two-stream methods could achieve high accuracy in activity recognition, they are not efficient to use because they take long time and huge computation resources to extract the optical flows in the videos.

The deep 3D CNN methods recognized the activities in the videos using a neural network that can learn the spatiotemporal features of the video from RGB frames. Tran et al. [31] first proposed the deep 3D CNN architecture, which used the 3D convolutional kernels to capture both temporal and spatial information from several consecutive frames. Then, the model was trained on a large-scale video dataset to recognize the activities. In terms of accuracy, this 3D CNN method was inferior to the original two-stream method [24], but it was much more efficient. Specifically, the 3D CNN method could process 672 frames per second, which is about 30 times faster than the two-stream method. Accordingly, numerous works have been done to improve the 3D CNN method. Hara et al. [32] created a 101-layer 3D CNN based on the residual network architecture [33], which verified that the deep 3D CNN could achieve the state-of-the-art performance in video activity recognition. Varol et al. [34] created a 3D CNN with the long-term temporal convolutions (LTC), which could capture the long-temporal features of the activity. Diba et al. [35] proposed a novel deep 3D CNN with variable 3D convolution kernel depths. This architecture was designed to concatenate temporal features-maps extracted at different temporal depth ranges, rather than fixed 3D homogeneous kernel depths, and it achieved good performance in both long and short video datasets.

3. Research challenges and objectives

The literature review indicated that existing vision-based methods for activity recognition in construction still have several limitations. Table 1 summarized these limitations and provided the comparison with the proposed method, which will be explained in Section 4. Specifically, the feature-based methods (e.g. [8,14,15]) could not recognize consecutive activities in long video sequences, since they have to pre-determine the start and finish of each activity. In addition, the performance of the feature-based methods in the long video sequence was not good, because the drift errors caused by trajectories of motion features accumulated with the progress of the time. Moreover, most feature-based methods just focused on monitoring single equipment. They focused on activity classification and barely considered the integration of detection and tracking technologies, which are necessary for multi-equipment activity recognition.

The detection-tracking-based methods (e.g. [16,17] [19–21]) also have the following limitations. They could just classify the activities into working and idling states, which provides limited information for productivity calculation and operation analysis. They estimated the activity based on the location relationships between excavators and trucks in video frames. Therefore, they cannot ascertain whether an excavator is truly active or idle. The detection-tracking-based methods need to pre-define thresholds and adjust them with the change of cameras positions, which is inconvenient to use. Instead of using the 3D physical distance, these methods estimated activities based on the 2D pixel distance between excavators and trucks. However, most images are taken within an inclined angle in the construction site. Therefore,

Table 1
Overview of gap-in-knowledge for equipment operation monitoring.

Category	Reference	Method	Test accuracy ^a (%)	Long video	Multiple excavators	Detailed activities	Cycle time	Automated productivity calculation	Limitations
Feature-based approach	Gong et al. [11]	3D-Harris feature + Bayesian learning classifier	79.00			✓			1. Pre-determine the starting point and duration of each activity
	Yang et al. [8]	HOG, HOF, MBH features + SVM classifier	57.00			✓			2. Single equipment
	Golparvar-Fard et al. [15]	3D HOG feature + SVM classifier	86.33			✓			3. Short video clips
Detection-tracking-based approach	Zou and Kim [16]	Color space detector + Centroid distance tracker	97.70	✓			✓		1. Activities not in detailed categories
	Bigler et al. [17]	GMM detector + Kernel covariance tracker	71.20	✓	✓		✓		2. Cannot distinguish whether the activity is truly acting
	Kim et al. [19]	HOG-based detector + Tracking-Learning-Detection tracker	91.27	✓	✓				3. Pre-define and adjust the threshold
	Azar et al. [7]	HOG-based detectors + Mean-shift tracker	95.00	✓			✓		4. Distance compaction influence
	Kim et al. [20]	Region-based CNN detector	99.99	✓			✓		5. Multiple excavators
Hybrid approach	Luo et al. [21]	Faster R-CNN detector + Relevance network	80.50		✓	✓			
	Kim et al. [23]	Faster R-CNN detector + Tracking-Learning-Detection tracker + CNN&LSTM classifier	90.90			✓	✓		1. Not applicable on long videos
	proposed framework	Faster R-CNN detector + Deep SORT tracker + 3D ResNet classifier (spatial-temporal-feature)	87.60	✓ (92.5%)	✓	✓	✓	✓	2. Activities have sequential relationships

^a All the methods were tested with different datasets.

the 2D distance is shrunk and cannot reflect the real spatial relationship between excavators and trucks. Existing detection-tracking-based methods estimated excavators' activities depending on their relative locations with trucks. If several excavators and trucks appear in the same video frame, it will be hard to build spatial correlations between the excavators and trucks. It should be noted that the accuracy of the methods listed in the Table 1 are tested on different datasets, which are not publicly available.

This research will address the limitations in the abovementioned research works. The overall objective of this study is to propose a solution to automatically monitor the operation and calculate the productivity of multiple excavators. In this study, a framework, which integrates both detection and tracking techniques is developed to monitor the operations of multiple excavators. In addition, a 3D CNN method is designed to precisely recognize excavators' activities (e. g. digging, loading and swinging) in video sequences. Finally, an algorithm is developed to automatically analyze the operation time and calculate the productivity of the excavators.

4. Proposed framework

The framework for multiple excavators' activity recognition and productivity analysis is shown in Fig. 1. The framework contains five main modules: excavator detection, excavator tracking, idling state identification, activity recognition, and productivity analysis. First, a detector is used to identify all the excavators in video frames. The detection results provide two kinds of data, i.e. equipment type and region i. Second, each individual excavator and its trajectory are identified through the detection-based tracking. The tracking results return the identification (ID) number and bounding boxes for each excavator. By analyzing the changes of centroid coordinates and areas of bounding boxes of the excavators in consecutive frames, idling states are identified. Then, the activities of the tracked excavators are recognized with a spatio-temporal feature-based 3D CNN model. Based on the activity recognition and idling state identification results, each video frame is labeled accordingly. Finally, the productivity of each excavator is calculated by compiling the activity recognition results. The details about each module in the framework are provided in the following sub-sections.

4.1. Excavator detection

The purpose of this module is to extract the excavators in video sequences. An accurate detection method is used here to detect multiple excavators in long construction site surveillance videos. Construction sites have complex working environments, which include various movements of workers and equipment. In order to isolate each excavator, the excavator detector is used to precisely get the regions of the excavators in video frames for the activity recognition use.

In this paper, we used a Faster R-CNN [22] model to detect the excavators in video frames. Faster R-CNN is selected as it has a high accuracy and low processing time. Additionally, experimental results of numerous research works have verified the high performance of Faster R-CNN in various construction objects detection [21,30,36]. The Faster R-CNN method includes three main steps. First, a CNN is applied to extract features of the whole image and produce a feature map. Then, a region proposal network uses the feature map to select 300 regions that are distinguished from background regions, and may include target equipment. Finally, the proposed regions are converted to 7*7*512 dimension feature vectors and passed to the fully connected layer to classify whether the region is an excavator or not. When the video is taken as the input of the Faster R-CNN model, it will return the type of the equipment and rectangular bounding boxes, which indicate the regions of the objects in video frames. The coordinates of bounding boxes from the detection results will be used as the input of the next tracking module.

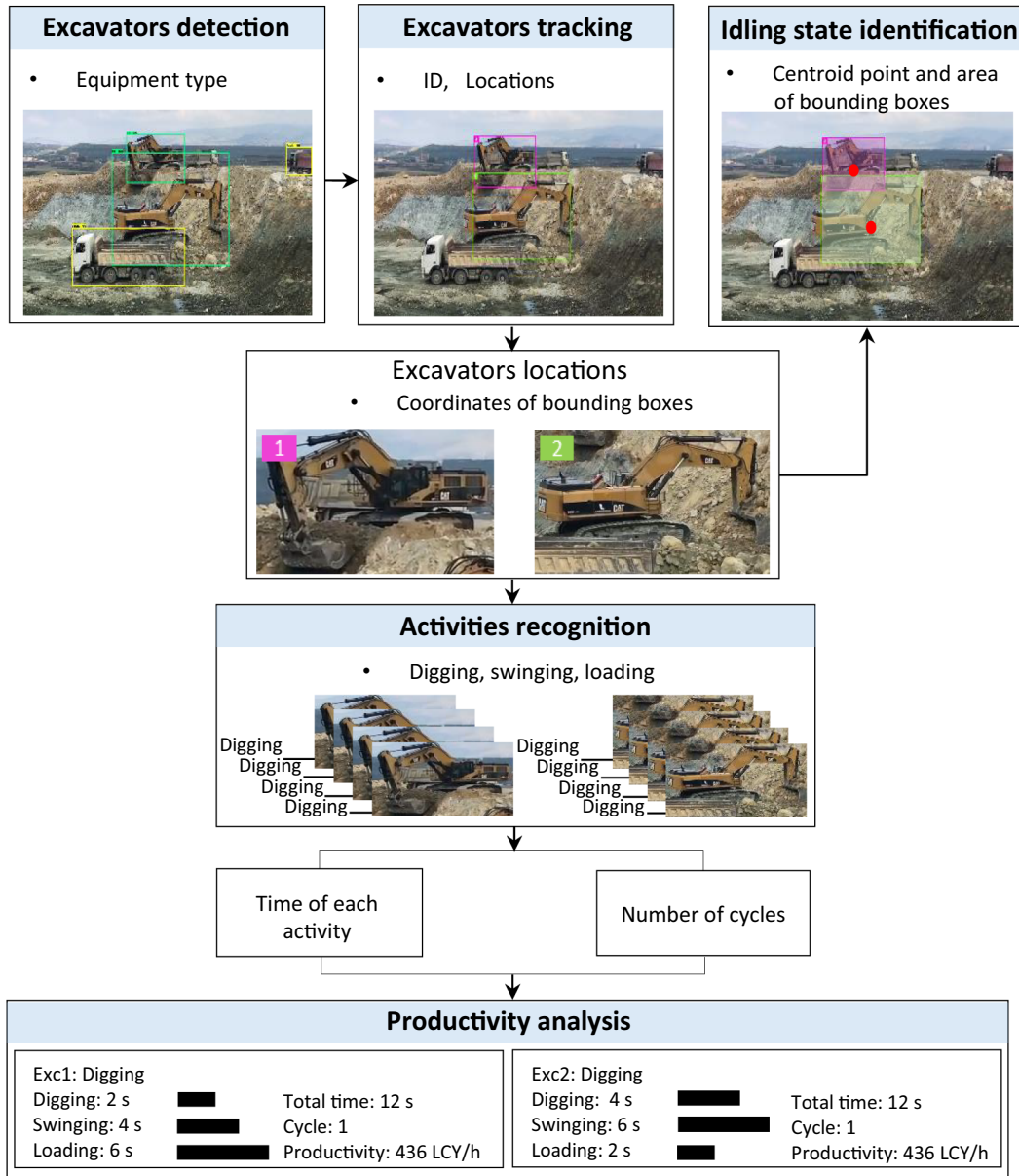


Fig. 1. Workflow of the proposed framework.

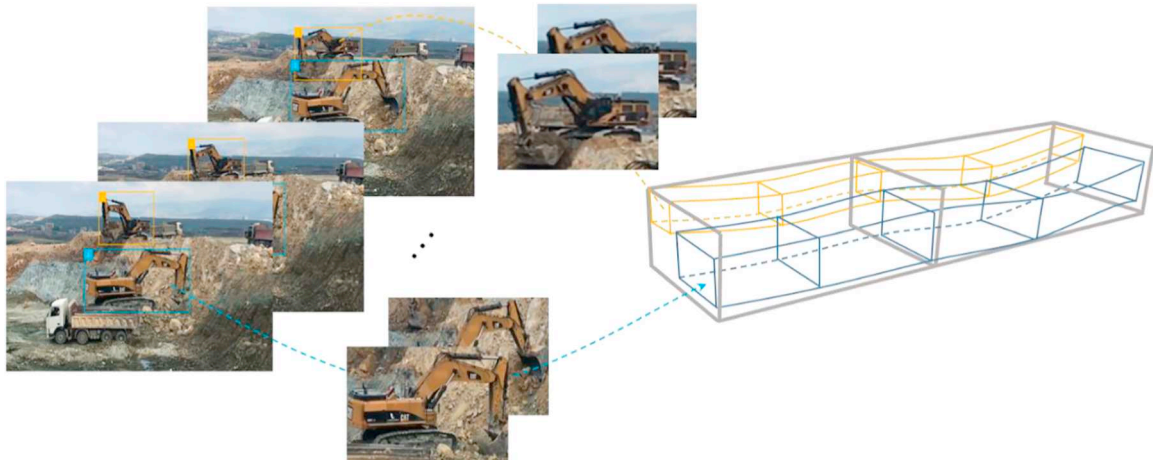


Fig. 2. Tracking results schematic plot.

4.2. Excavator tracking

In this module, the most state-of-the-art multi-object deep Simple Online and Real-Time (SORT) tracker is applied to associate the same excavators detected in the previous step across all the frames in the video [37]. The deep SORT tracker is selected since it achieved good performance on MOT 16 dataset, and it could track the objects through long periods of occlusions in the video. The deep SORT tracker uses both appearance and trajectory information provided by the detection results to track the excavators in video frames. In the trajectory aspect, the tracker uses a constant velocity model [38] to predict the bounding box of the target excavator in the current frame based on the previous ones. In the appearance aspect, a ten-layer CNN model is used to extract the appearance features of the region inside the bounding box, and represent it with 128-dimensional vectors. Finally, the detected excavator in the current frame is related to the predicted bounding boxes with the Hungarian algorithm [39]. As a result, the same excavator is detected and matched in different frames. The region for the detected and matched excavator is further extracted and resized to 112×112 pixel and input to the activity recognition module. The tracking results could be considered as several activity tubes, as shown in Fig. 2. Each tube represents the exact regions of a certain excavator across the video frames, which is the basis of the next step.

4.3. Idling state identification

The purpose of this module is to identify the idling state of the excavators. The workflow of idling state identification is shown in Fig. 3. First, the pixel coordinates of all detection/tracking bounding boxes in M frames of the video are extracted. The centroids (x_i, y_i) and areas (a_i) of the bounding boxes are calculated. Also, the average area $\bar{a}$ for all the bounding boxes is estimated. For each video frame, the centroid distance changes (Δd_i) and area change (Δa_i) between the current frame and the n^{th} frame after the current one are calculated using Eqs. (1) and (2);

$$\text{Centroid coordinates change } (\Delta d_i) = \sqrt{(y_{i+n} - y_i)^2 + (x_{i+n} - x_i)^2} \quad (1)$$

$$\text{Area change } (\Delta a_i) = |a_{i+n} - a_i| \quad (2)$$

where (x_i, y_i) and a_i are the centroid coordinates and area of the bounding box in frame i . Moreover, a sliding window with n consecutive frames and the step of one frame is applied to calculate the standard deviation of distance changes $STD(\Delta d_i)$ and the standard deviation of area changes $STD(\Delta a_i)$. Figs. 4 (a) and (b) show an example of changes of distance (Δd) and the concept of sliding window for calculating $STD(\Delta d)$, respectively. If $STD(\Delta d_i)$ is less than d' , and $STD(\Delta a_i)$ is less than $\alpha \bar{a}$, the excavator's state in that particular frame will be identified as idling. The threshold d' represents a negligible change of the distance between the centroids, and α is a percentage of the average bounding box area so that $\alpha \bar{a}$ represents a negligible change of the area. Both d' and α are determined based on the resolution of the video and the distance of the excavator from the camera.

4.4. Activity recognition

In the activity recognition task, the latest 3D residual neural network (3D ResNet) [32] is used to recognize the excavator's activities (e.g. digging, loading, and swinging) in video sequences. The 3D ResNet is selected since it achieved the best performance of activity recognition task on UCF-101 dataset [40] with 86.5% accuracy, and it could directly get the activity recognition results from RGB input video frames. The model has the same residual block architecture as the ResNet [37] but performs the convolution and pooling with 3D kernels. It preserves both the spatial and the temporal information of the activities in the video. All the inputs, kernels, and outputs in the network are 3D tensors with the dimensions of temporal $\times$ height $\times$ width ($L \times H \times W$) [31].

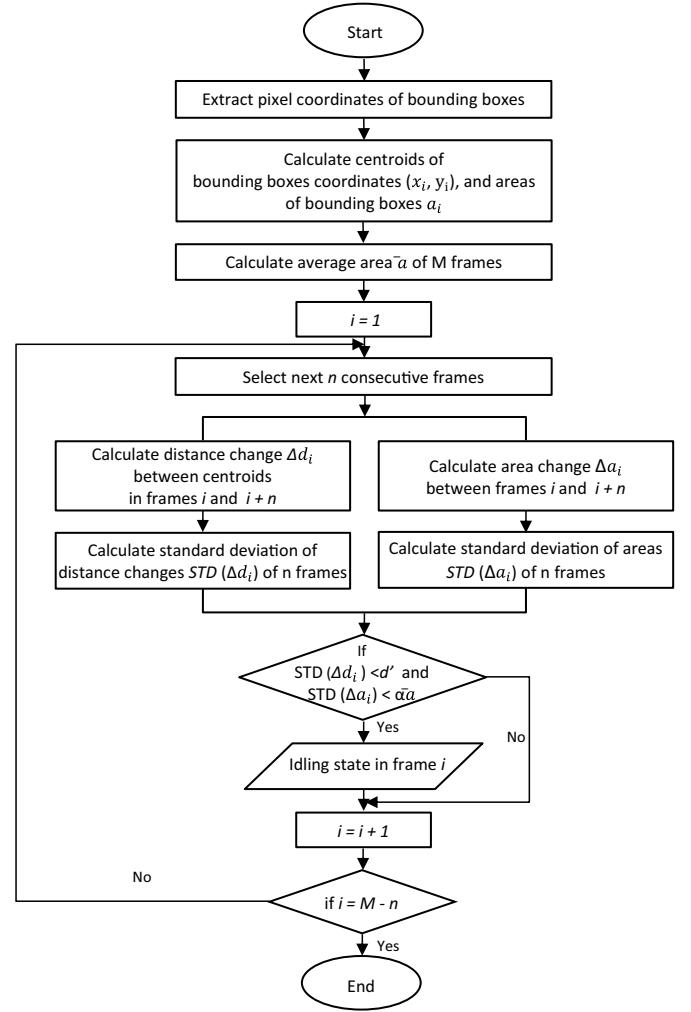


Fig. 3. Method of idling state identification.

Specifically, the network takes $16 \times 112 \times 112$ video frames as input. The sizes of the convolutional kernels are $3 \times 3 \times 3$, and the temporal strides are 1 for the first convolutional layer and 2 for the other layers. First, the tracking results are taken as the input of the neural network. Accordingly, convolutional kernels are applied on every 16 frames to extract the spatiotemporal information of the activity. The selection of 16 frames is mainly due to the GPU memory limit, although the selection of a longer duration may improve the recognition accuracy of the model [41,42]. At last, each frame is labeled to indicate the excavator activity after correcting the recognition errors with majority voting, as shown in Fig. 5. This is because excavators usually work continuously and cannot change their activity states in a short period of time. It was observed that each activity lasts at least 2 s during the operation.

The training of the activity recognition model started from fine-tuning the 3D ResNet work developed by Kay et al. [43]. The fine-tuning is to efficiently initialize the parameters of the model and avoid overfitting. In the training process, the data augmentation technique is used to extend the size of the dataset by flipping the video frames, shifting the image channels, and shearing the frame size. In addition, all video clips are fixed at 25 FPS for training, the batch size of the model was set to 16, and the learning rate was set to 0.001.

4.5. Productivity calculation

In this module, a method is created to automatically calculate the

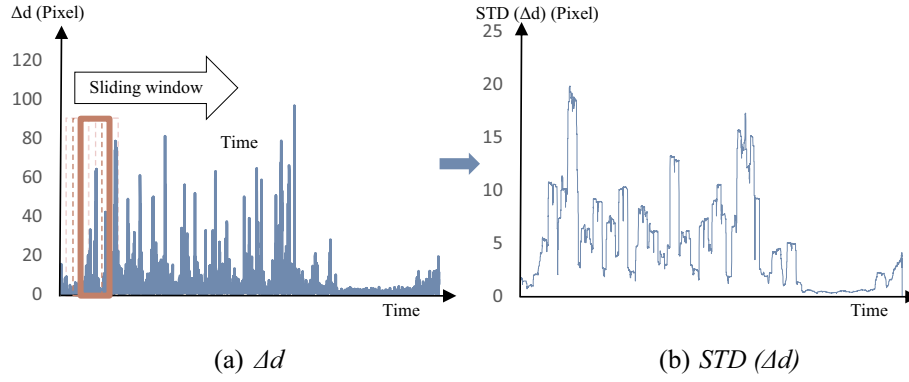


Fig. 4. Examples of changes of distance (Δd) between centroids of bounding boxes and the concept of sliding window for calculating $STD(\Delta d)$.

productivity of the excavator based on the activity recognition results. In the earthmoving work, excavators usually work with other equipment, such as trucks and scrapers. For example, an excavator digs the soil and loads it onto the bed of a truck. When fully loaded, the truck moves the soil to the dumping area and returns to be loaded again. If the excavator is considered as an independent equipment, its working process could be broken down into several activities such as soil digging, swinging and loading the truck. Therefore, a single earthmoving cycle of an excavator could be broken down into four types of activities: digging, swinging after digging, loading into the dump truck, and swinging for digging. During the work time, these four types of activities are repeated sequentially, which is defined as a production cycle, as shown in Fig. 6.

In earthmoving, the excavator's productivity can be calculated with the cycle time and the bucket payload, as shown in Eq. (3). Since the bucket payload is given by the manufacturer, the target of the productivity calculation becomes to determine the cycle time of the excavator. To simplify the procedure, the two types of swinging (swinging after digging and swinging for digging) are not distinguished in this paper. This way, one excavator working cycle is broken down into digging, swinging, and loading.

$$Productivity (LCY/hr) = \frac{Cycles}{hr} \times \frac{Average\ bucket\ payload\ (LCY)}{Cycle} \quad (3)$$

The time for each cycle is measured following the workflow in Fig. 7. After the activity recognition, each video frame is labeled to indicate the activity of the excavator in the frame. The labels are shown as L_i in Fig. 7. Then, the labels of two consecutive frames are compared. If they are the same, it means that the activity continues. Therefore, the time for this activity is increased by 1/FPS (frame per second). If the labels are different, it means that a new activity has started. The time of the newly recognized activity will increase by 1/FPS. The total time of one cycle is the difference between the start times of two adjacent digging activities.

5. Implementation and results

In this section, the implementation of the proposed framework is introduced. The activity recognition model is trained in the Python 3.6 environment, 64-bit Ubuntu 16.04 system with the hardware configuration of an NVIDIA GeForce GTX 1080GPU and 32 gigabytes. The

proposed framework is tested in the Python 3.6 environment, 64-bit Windows 10 system with the hardware configuration of an NVIDIA GeForce GTX 1070GPU and 32 gigabytes. It could process 15 frames per second with this configuration.

5.1. Training and testing

First, a dataset of videos containing excavators is manually collected. Each video contains one of the three types of activities: digging, loading and swinging. In order to avoid bias, 351 video clips were collected from 21 different construction sites, considering site conditions, equipment viewpoints, and scales and colors of excavators. The detailed information of the dataset is shown in Table 2. First, each video sample in the dataset is annotated with the activities of this video. Fig. 8 illustrates the example images extracted from the videos in the dataset.

Then, 30 video clips with the resolution of 1280×720 pixels and the duration of 264 s were used to test the performance of the activity recognition model. Each video clip in the test dataset contains at least one type of excavator's activity. Two indicators, precision and recall, were applied to measure the performance of the activity recognition model as shown in Eqs. (4) and (5). These two indicators are widely used to validate the performance of object classification and detection methods in the computer vision domain.

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

where TP is the number of true positives, FP is the number of false positives, and FN is the number of false negatives. More details about the concepts of precision and recall could be found in the work of Kim et al. [19]. The comparison of activity recognition results with the ground truth is shown in Fig. 9, and the performance of the pre-trained model is presented in Table 3. The recognition precisions for the three activities are: 95% for digging, 86% for swinging, and 84% for loading. The recall rates are 86%, 93%, and 80%, respectively. The average accuracy of the recognition is 87.6%. The high activity recognition accuracy on test videos, which have various viewpoints, background noises from other movements, and partial occlusions, validated the robustness of the proposed method.

In addition, it is worth noting that the model was also applied on a

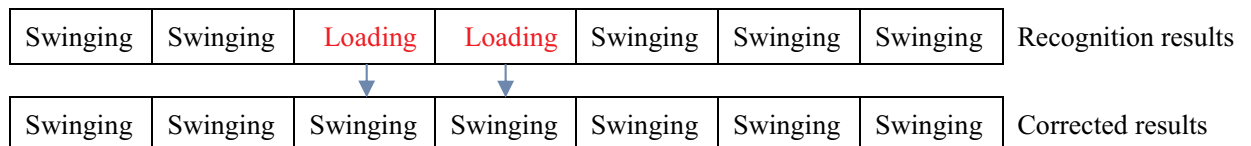


Fig. 5. Correction of abnormal activity recognition.

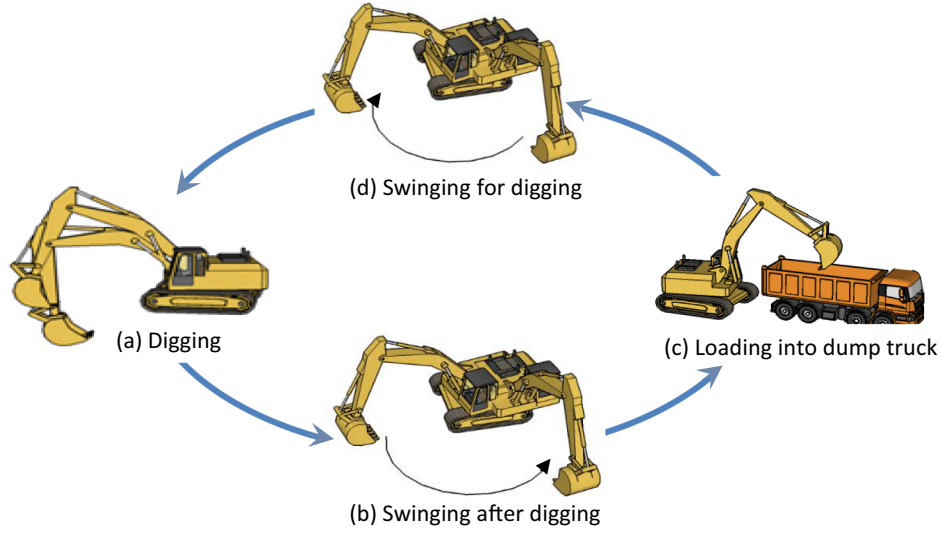


Fig. 6. Excavator working cycle.

60.2 min video to recognize excavator's activities in the implementation stage and achieved the accuracy of 92.5%. The results indicate that the proposed method can effectively recognize the consecutive activities of the excavator in the long video sequences.

5.2. Activity recognition of multiple excavators

In this section, two case studies were conducted to validate the feasibility of the proposed method for multiple excavators' activity recognition.

5.2.1. Case 1: Two excavators without overlapping

In Case 1, the proposed method was used to recognize two excavators' activities in a 1 min video (Fig. 10). In this video, Excavator 1 has 8 activities within 2 cycles, and Excavator 2 has 10 activities within 2.5 cycles. Fig. 11 shows examples of the activity recognition. The test results are shown in Table 4.

Table 2

Statistic information of the dataset.

Activity type	Number of videos	Total time (s)	Average of video length (s)	Number of excavators
Digging	122	651	5.3	19
Swinging	119	440	3.7	21
Loading	110	490	4.5	19

The results show that there are four wrongly recognized activities: three for Excavator 1 and one for Excavator 2. In fact, the errors are mainly due to the bad light conditions as can be seen in Fig. 8. The light condition changes drastically in the second half of the video.

5.2.2. Case 2: Two excavators with overlapping

In Case 2, the proposed method was used to recognize two excavators' activities in a 40 s video (Fig. 12). Excavator 1 has five

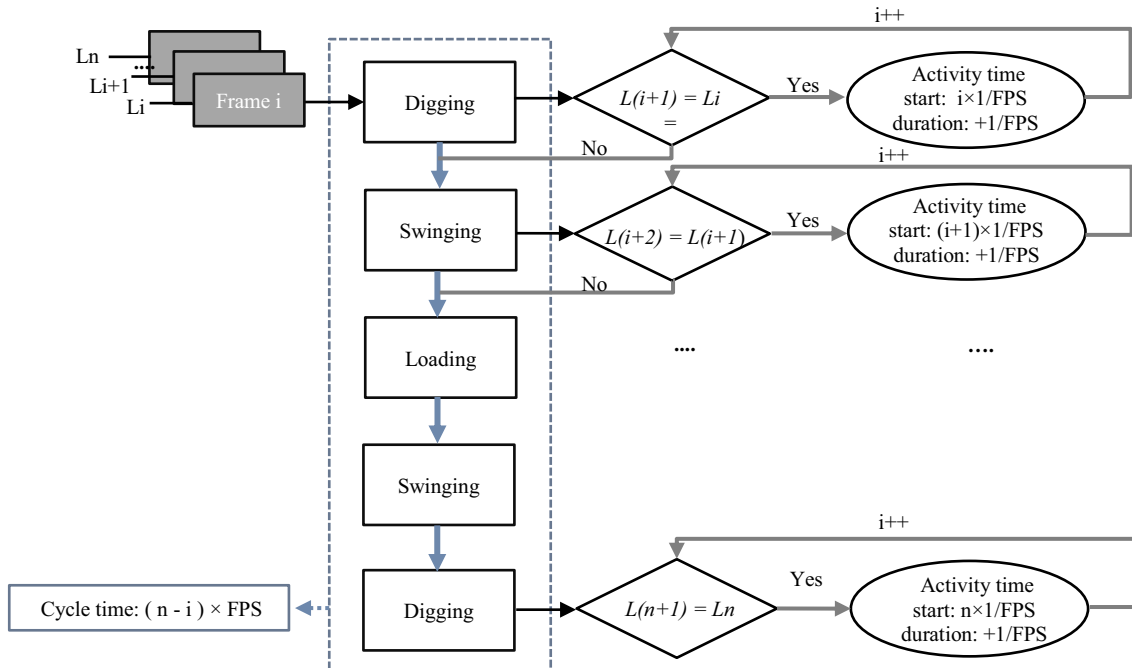


Fig. 7. Workflow of cycle time calculation.

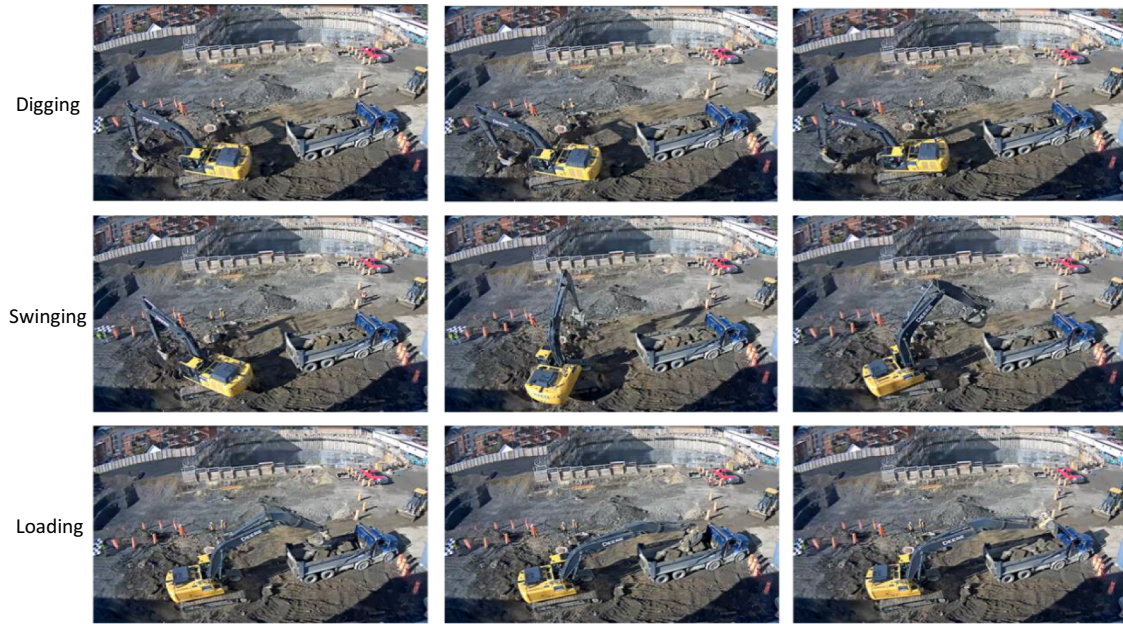


Fig. 8. Example images extracted from the dataset.

activities within 1.25 cycle, and the Excavator 2 has six activities within 1.5 cycle. Fig. 13 shows examples of the activity recognition. The test results are shown in Table 5.

The results indicate that one activity is wrongly recognized for Excavator 1 because of the overlapping of the bounding boxes as shown in Fig. 12. The deep SORT tracker. Identified a large bounding box for Excavator 1 fully overlapping with Excavator 2 in 500 frames, which affected the activity recognition results.

5.3. Productivity estimation

Additionally, the proposed framework was also tested to estimate the productivity of excavators using a long video sequence, which contains 60.2 min of excavator's operation. Based on the method of idling state identification explained in Section 4.3, the width of the

sliding window n is selected as 100 frames (equivalent to 4 s with 25 FPS video). The threshold d' is selected as 1 pixel, and α is selected as 1.5% considering possible small changes in the bounding box even when the excavator is in idling state. The test outputs are shown in Fig. 14. The activity information is added on bottom of the image including: (1) the activity identified in the current frame; and (2) the total time of each activity compiled till the current frame. In addition, the last frame provides the productivity information, which include: (1) the total time of the video; (2) the number of working cycles; and (3) the productivity of the excavator with the unit of LCY/h.

In the video, the Deere 290G excavator (bucket payload of 1.4 LCY) completed 94 work cycles in 60.2 min with the idling time of 9.7 min and the actual productivity of 131.16 LCY/h. The estimated total idling state duration is 9.1 min, and the estimated productivity is 153.48 LCY/h with 110 cycles. The accuracy of productivity calculation is 83%. The

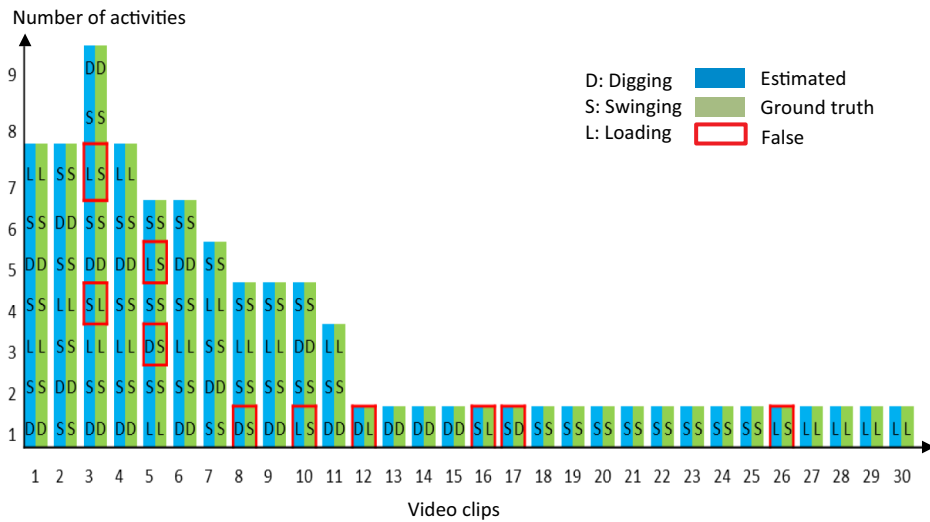


Fig. 9. Comparison of test results and ground truth values.

Table 3
Activity recognition results.

Activity	Number of activities in the videos		Performance metrics		
	Total	Incorrect recognition	Precision (%)	Recall (%)	Accuracy (%)
Digging	21	3	95	86	87.6
Swinging	40	3	86	93	
Loading	20	4	84	80	
Average/total	81	10	88	88	

test results showed the feasibility of using the developed framework to analyze real videos of construction projects and to monitor the operation of excavators.

6. Summary, contributions and future work

6.1. Summary

This paper presented a novel fully automatic vision-based framework for multiple excavators' operation monitoring and productivity calculation. The framework integrates the detection, tracking, activity recognition and productivity calculation modules. The detection module identifies all the excavators in video frames. The tracking module correlates the same excavators and extracts their regions across the frames in the video sequences. After detecting the idling state, the activities of each excavator are recognized based on the tracking results. Finally, the productivity is automatically calculated based on the activity information of the excavators. The effectiveness of the proposed framework has been tested on construction surveillance videos, and the results have shown the feasibility of the proposed framework.

6.2. Contribution and conclusions

The contributions of this work mainly consist of three perspectives. First, the advanced technology in computer vision is integrated to recognize continuous activities of excavators in long site surveillance videos, as shown in Table 1. Specifically, compared with the recent work of Kim et al. [19,20] which could just identify working and idling states of the excavator based on its relative location with respect to the

truck, the proposed method can recognize detailed activities such as digging, loading and swinging from activities spatial and temporal features. In addition, the proposed method is also superior to another related work of Kim et al. [23] in long videos analysis, since they did not consider the temporal features of the activities. In Kim et al.'s method, activities are recognized based on their sequence relations. The method in this paper can recognize activities based on both spatial and temporal features that are directly learned from videos regardless of their sequential relations. In the model training stage, the proposed method relied on several training strategies, such as fine-tuning, data augmentation and batch size optimization, to train the detection, tracking and activity recognition models with a limited dataset. The results of the activity recognition on different excavator sizes and types, various viewpoints and lighting conditions, and background movements have proved its robustness in construction environments and provided a solid basis to automate the calculation of the equipment productivity.

Second, the detection, tracking and recognition techniques are integrated to measure multiple excavators' operations. In a recent construction activity recognition research, Luo et al. [29] manually specified bounding boxes to start the tracking process, which is not efficient. In this paper, a fully automated framework is conducted for excavators' activity recognition without the manual input.

Third, a sliding window method is applied to identify the idling state of excavators by comparing the changes of centroid distance and areas of bounding boxes of excavators in consecutive video frames.

Forth, an automatic method is developed to directly analyze the productivity based on the activity recognition results. The experimental results on the video with the duration of nearly one hour supported the



Fig. 10. Activity recognition under contrast lighting conditions in Case 1.

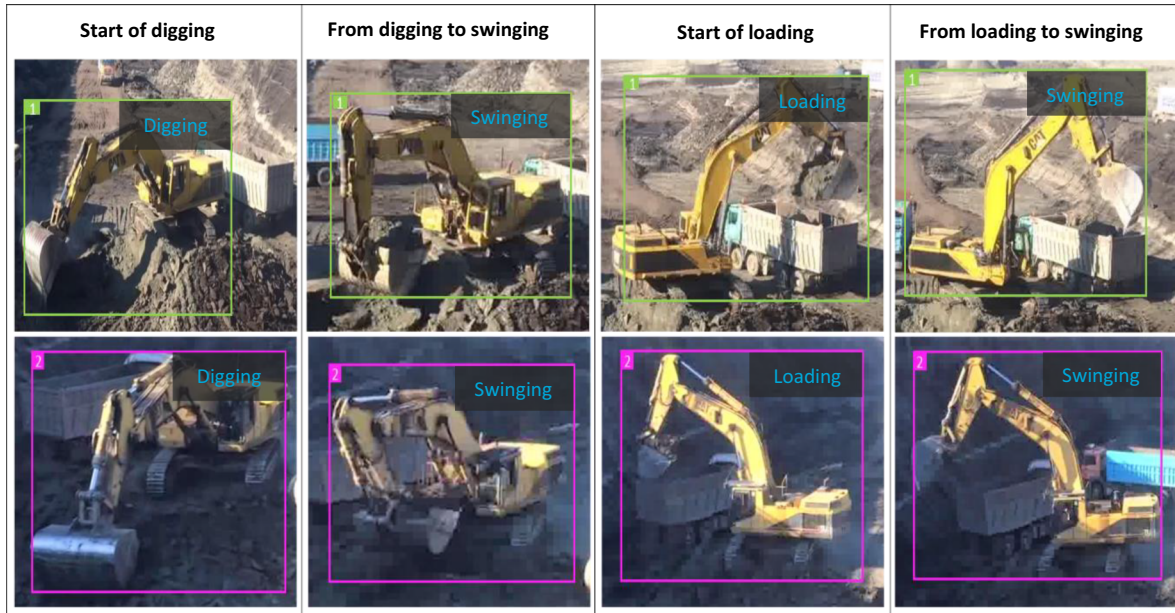


Fig. 11. Examples of activity recognition in Case 1.

Table 4

Test results and ground truth values of Case 1.

Excavator 1	Cycle 1	Ground truth	D	S	L	S
		Estimation	D	S/D	L	S
	Cycle 2	Ground truth	D	S	L	S
		Estimation	D/L	S	D	S
Excavator 2	Cycle 1	Ground truth	D	S	L	S
		Estimation	D	S	L	S
	Cycle 2	Ground truth	D	S	L	S
		Estimation	D	S	L	S
	Cycle 3 (partial)	Ground truth	D	S		
		Estimation	D/L	S		

D: Digging; L: Loading; S: Swinging.

feasibility and applicability of the proposed method in practice. Moreover, the detailed operation information provided by this method, such as the time of each activity and the number of work cycles can be

further used to help construction managers identify the causes of excavator's low productivity from the video.

6.3. Limitations and future work

There are several limitations in this research: (1) The activity recognition performance is affected by the detection and tracking results. When the bounding boxes of two excavators are fully overlapping in the detection and tracking results, the activity of one excavator maybe affected by the other excavator as shown in Fig. 10; (2) The light condition of the video also affects the activity recognition result. When the light is too bright or too dark, it is difficult to recognize the moving features in video frames, as shown in Fig. 8; (3) It is noted that the diversity of our dataset is limited, which have negative influence on both recognition and productivity results. For instance, some of the videos in our dataset are taken from low heights. However, most of the construction surveillance videos are overlooking from high points.

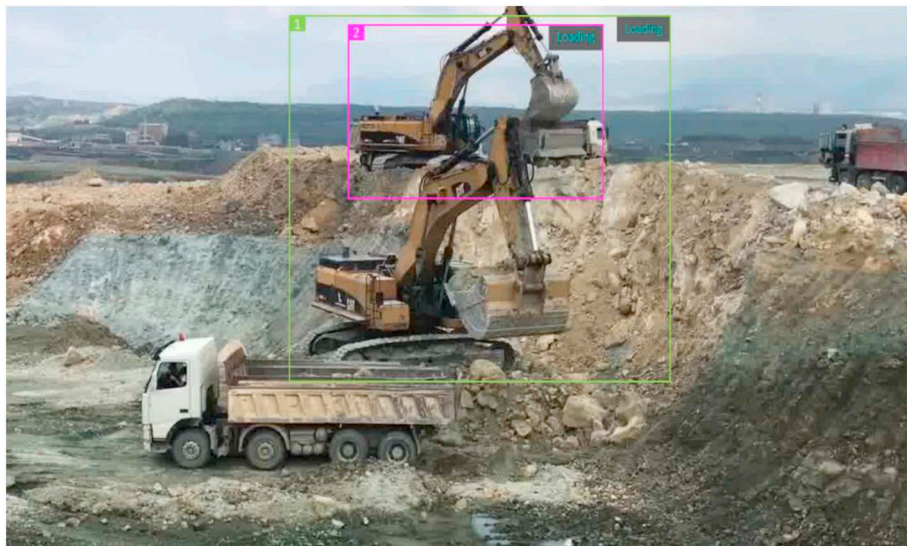


Fig. 12. Activity recognition with overlapping bounding boxes in Case 2.

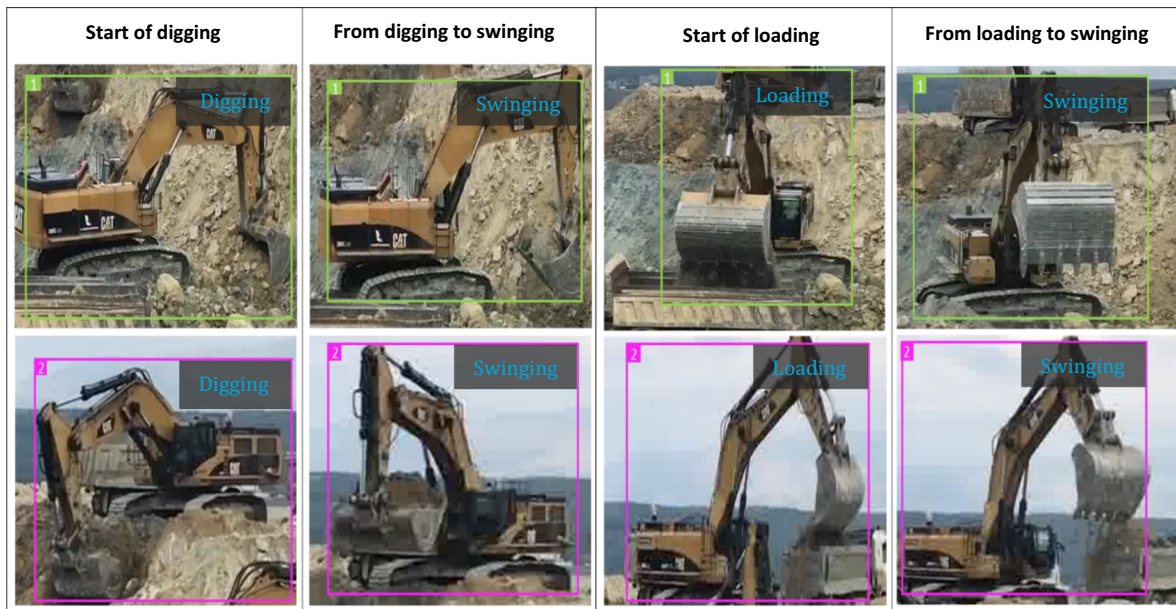


Fig. 13. Examples of activity recognition in Case 2.

Table 5

Test results and ground truth values of Case 2.

Excavator 1	Cycle 1	Ground truth	D	S	L	S
		Estimation	D	S	L	L
	Cycle 2 (partial)	Ground truth	D			
		Estimation	D			
Excavator 2	Cycle 1	Ground truth	D	S	L	S
		Estimation	D	S	L	S
	Cycle 2 (partial)	Ground truth	D	S		
		Estimation	D	S		

D: Digging; L: Loading; S: Swinging.

When considering the application of the research, the previous limitations have to be solved in future work: (1) To improve the robustness of the method, such as, avoiding the error caused by the failure

of detection and tracking or bounding boxes occlusion, videos taken from more than one camera should be aligned and applied to recognize the activities. Moreover, to avoid the influence of the bad light condition, the video should be preprocessed by adding a filter; (2) In order to train a better performance model for activity recognition, various data collected from the construction site surveillance video should be added to the dataset; (3) To improve the efficiency and optimize excavator's productivity, the activities of the truck that works with the excavator should also be recognized. By analyzing the relationship of excavator's cycle time with the truck's position, excavator's productivity can be optimized (e.g. by improving the maneuvering). Moreover, by analyzing the cycle time of the truck, the number of trucks are required to keep the excavator working at full capacity can be calculated. This could be done based on identifying the work groups of excavators and trucks; (4) For the purpose of productivity calculation, having long

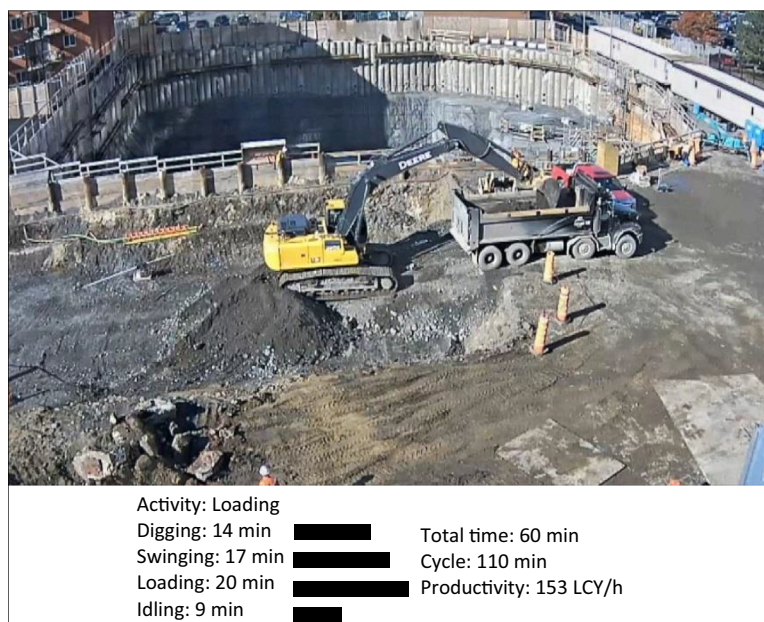


Fig. 14. Examples of productivity calculation results.

videos (i.e. about an hour) is important to capture the variation in cycle times. However, the process of validation long videos is time and labor intensive. Therefore, more long videos will be tested in the future.

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Author declaration

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